

The Intraday Seasonality of Market Trends

When Is the Market Most Likely to Trend Intraday?



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In this article, we study the **intraday seasonality of trend-following strategies on S&P 500**.

In simple terms, we want to understand **when, during the trading session, the S&P 500 is more likely to generate trend-following profits, and whether this pattern has changed over time**.

This study was motivated by questions from our readers, who noticed, at least qualitatively, that SPY seemed to trend less consistently during certain parts of the trading session.

The goal of this article is to **test this hypothesis empirically** using our database and an **ensemble of six different intraday trend-following models**.

We begin by introducing the models. We then construct an ensemble portfolio to reduce model-specific noise in the results. Finally, we evaluate intraday seasonality over the full sample and investigate whether it has changed in recent years.

Preview of the results: Intraday trend-following profits have not disappeared, but they have weakened during certain parts of the trading session.



Intraday Trend-Following

Over the past few years, we have published several empirical studies on intraday trend-following strategies in U.S. equity markets.

Many of these ideas were first introduced by traders such as **Toby Crabel, Linda Raschke, Larry Connors, Monroe Trout**, and others in the early '90s. The basic intuition is simple: when price moves significantly away from a reference price level during the intraday session, the move may reflect a **genuine imbalance between supply and demand**.

Given our belief that capital can only move gradually through the market, we think **these imbalances can persist throughout the trading session**, allowing intraday trend traders to capitalize on them.

Our main **contribution** compared with previous studies is the **accuracy** and **transparency** of the empirical analysis. Previous studies relied on daily data and did not include explicit position management rules. In our papers, we used intraday data, incorporated simple position management rules, adopted realistic trading frequencies, and, in many cases, also shared the Python code we used to run the backtests.

For readers interested in the background, three useful references are our previous articles: *Beat the Market*, *Can Day Trading Really Be Profitable?*, and *VWAP: The Holy Grail for Day Trading Systems*.

In this article, we use an **ensemble of intraday trend-following models** that are very similar to the type of models we use in our own portfolios.

The reason we use an ensemble of models is simple: if we study only one model, the result may be driven by that model's own rules. On the other hand, if we **combine several models that all try to capture intraday trend continuation in slightly different ways, we have a better chance of isolating the underlying market behavior.**

An Ensemble of Models

We use six intraday trend models. All of them trade SPY, the most liquid ETF tracking the S&P 500.

Below is a description of each model used in the study.

- **Model #1. SPY breakout model using bands calibrated on 10-day realized daily volatility.** The upper band is built around the higher of today's open and yesterday's close, while the lower band is built around the lower of the two. The model starts trading after the first 30 minutes, checks signals every 30 minutes using staggered rebalance clocks, goes long above the upper band and short below the lower band, and exits when price moves back inside the band or at the close.
- **Model #2.** SPY breakout model with the same entry logic as Model #1, but with a **different exit reference**. Long positions are closed when price falls back below the higher of today's open and yesterday's close; short positions are closed when price rises back above the lower of the two. Signals are checked every 30 minutes using the same staggered clocks.
- **Model #3. SPY trend model inspired by our Beat the Market paper.** It estimates a time-varying intraday noise area from the average absolute move from the open at each minute over the previous 14 days. The model trades breakouts only when price is also on the right side of VWAP, checks signals every 30 minutes,

and exits when price returns inside the noise area, crosses VWAP, or reaches the end of the day.

- **Model #4. SPY volatility-breakout model that checks signals every 5 minutes.** Its band width is based on a 21-day average of the daily intraday noise, measured as the average absolute move from the open during the session. In this study we remove the original volatility-contraction day filter, so the model is allowed to trade every day. Exits are based on the session open, VWAP, or the close.
- **Model #5. SPY volatility-breakout model using 21-day realized daily volatility bands around the open and yesterday's close.** It checks signals every 30 minutes, uses staggered rebalance clocks in the ensemble, enters outside the bands, and exits when price moves back inside the relevant band or at the end of the day.
- **Model #6. SPY VWAP direction model inspired by our VWAP's paper.** It can rebalance every minute: long when price is above VWAP, short when price is below VWAP, and flat at the end of the day.

Four of these models operate on a 30-minute decision frequency. A naive implementation would check all of them at exactly the same timestamps, such as 10:00, 10:30, 11:00, and so on. This procedure can create artificial PnL spikes in the ensemble portfolio, as several models may enter or exit at the same time.

To **avoid this synchronization effect, we run each 30-minute model on six staggered grids** (i.e., :00/:30, :05/:35, :10/:40, :15/:45, :20/:50, and :25/:55). We then average these six versions within the same model before combining them with the rest of the ensemble.

All models are run gross of transaction costs, with a simple sizing approach. When a trend is long, exposure is fixed at +100%; when it is short, exposure is fixed at -100%. Positions are closed at the end of the day. We intentionally avoid adding sizing or transaction cost

assumptions because the objective here is simply to assess intraday seasonality.

How We Measure Intraday Trend PnL

For each model, each trading day, and each minute of the session, we compute the strategy PnL in basis points.

This gives us a large table with five columns. Below a snapshot from the real 1-minute panel.

model	day	time of day	PnL in bps	active_trends
Model #1	06-Jan-2026	11:26	-0.8	3
Model #2	06-Jan-2026	11:26	-5.1	3
Model #3	06-Jan-2026	11:26	0.0	3
Model #4	06-Jan-2026	11:26	0.0	3
Model #6	06-Jan-2026	11:26	5.1	3

Snapshot from the real 1-minute panel; PnL rounded to one decimal.

Note: A careful reader may notice that Model #1 shows a PnL of only -0.8 bps, while some other active models show about +/-5.1 bps. This is because the 30-minute models are run on six staggered rebalance clocks and then averaged inside the same model family. In this specific minute, only one of the six Model #1 variants was exposed, so its effective exposure was 1/6 and its PnL is roughly one sixth of the full SPY minute move.

The variable **active_trends** counts how many models have non-zero PnL during each minute of each trading day. We need this variable because we want to remove minutes when the entire ensemble is inactive,

meaning that no intraday trend was detected by any model. **If all models are flat, that minute does not provide any meaningful information about the time-of-day seasonality of intraday trends.**

At the same time, if only one model is active, we still include the zero-PnL observations from the other models in the calculation. This prevents isolated signals from receiving too much weight. A minute where only one model detects a trend should naturally carry less weight than a minute where several models agree on the same trend.

We then compute the **average PnL, in basis points, for each minute of the session across the full sample and all models.**

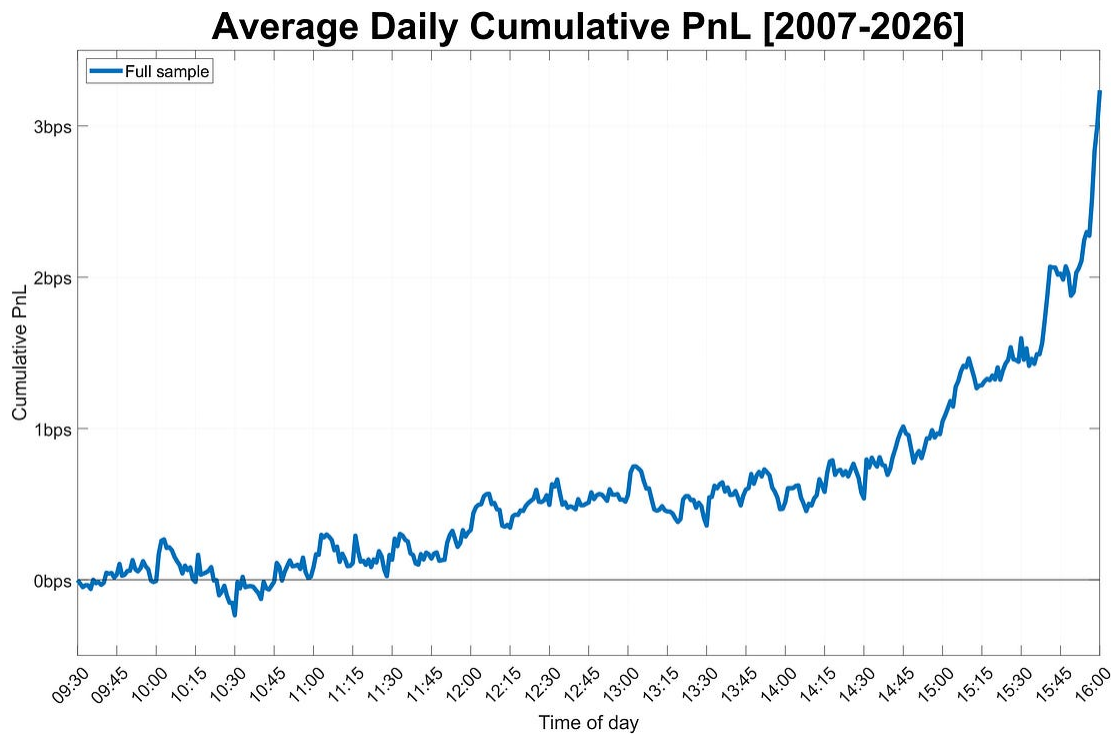
Finally, we calculate the cumulative sum of these average 1-minute PnL values.

time of day	average PnL in bps	cumulative PnL in bps
15:56	-0.02	2.27
15:57	0.24	2.51
15:58	0.32	2.84
15:59	0.16	3.00
16:00	0.24	3.24

Snapshot from the full-sample average-by-minute table; values in bps.

Results

The following charts shows the average cumulative intraday trend PnL for the full sample, from mid 2007 until July 2026.

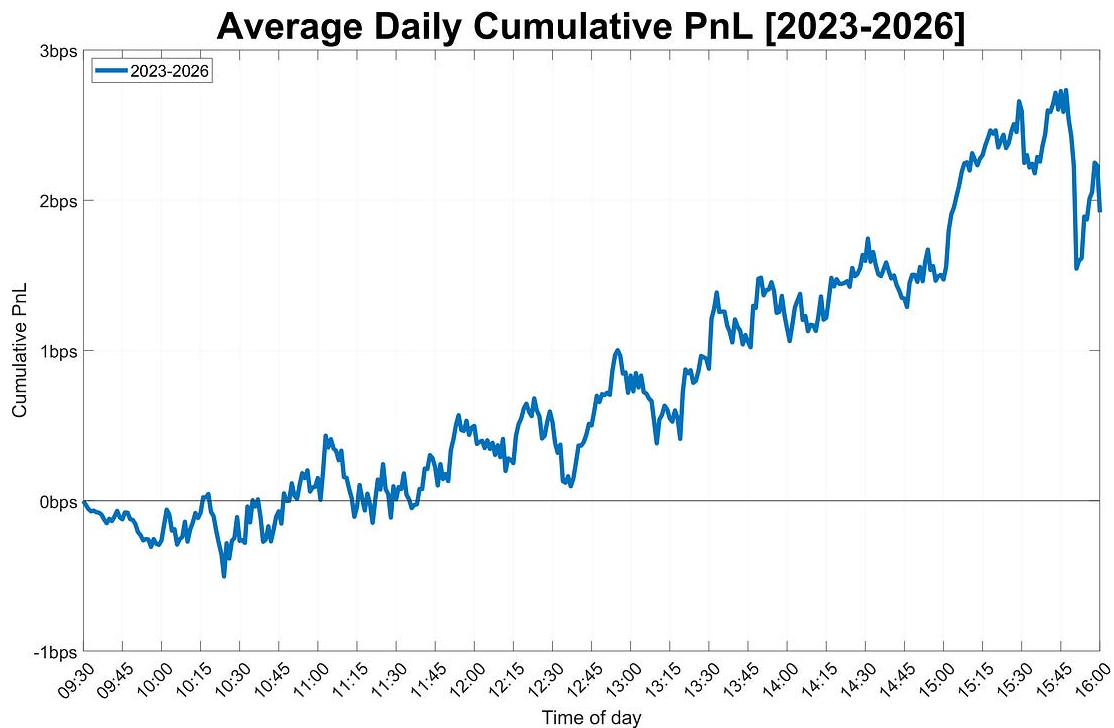


The full-sample analysis clearly **demonstrates that intraday trend-following profits accumulate gradually during the session and accelerate significantly into the close.**

The last hour contributes for roughly 60% of the full-day PnL, while during the last 30 minutes the cumulative PnL almost doubles.

We then repeat the same exercise only using the last four calendar years, **2023-2026**.

The same chart looks different.



As expected, given the shorter sample, the cumulative profile is now more volatile. Nevertheless, **the overall PnL trajectory still points upward**. After a soft start, the curve gradually drifts higher throughout the session, with a few visible pullbacks, and reaches a peak around 15:45.

The problem is what happens next...

Between 15:45 and 16:00, the curve gives back a significant portion of the afternoon gains and ends the day almost 30% below its maximum value. The overall result remains positive, but the shape is very different from the full-sample profile, **where the final minutes add to the trend rather than detract from it.**

We can't prove from this analysis that the market structure has permanently changed, but the results suggest that the old assumption that late-day trends are usually the cleanest part of the session deserves to be re-evaluated.

The Heatmap

The next figure shows the same analysis from a different perspective.

Each cell represents the **average 15-minute PnL** of the ensemble portfolio for a given year and time of day. Green cells indicate positive trend-following PnL, red cells indicate negative PnL, and white cells indicate periods with little or no average PnL.

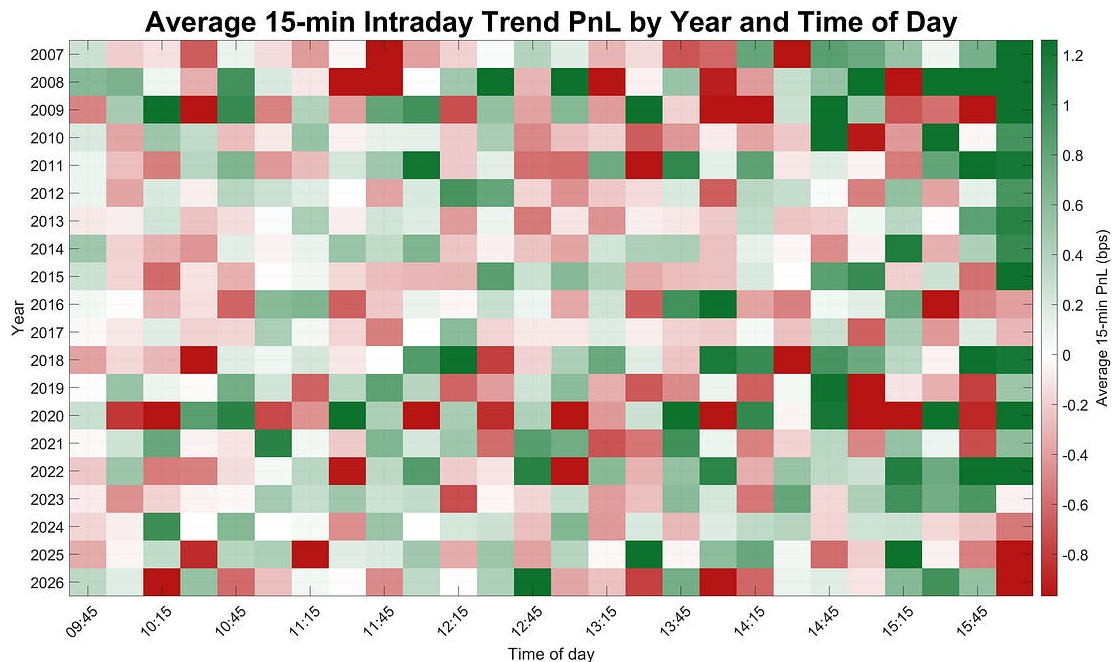


Figure 3. Average 15-minute intraday trend PnL by year and time of day. Positive values are green, negative values are red, and the color scale is clipped at the 5th and 95th percentiles.

The heatmap confirms the main point. Intraday trend seasonality is not constant through time. The early part of the sample shows broad trend continuation into the close, especially in the final 15 minutes, while in **recent years, instead, show a stronger tendency for trends to vanish or reverse near the end of the session.**

The Cause Behind Reversals?

At this stage, any explanation should be treated as a **hypothesis rather than a conclusion**.

The explanation we find most plausible is the rapid growth of ODTE options and the mechanical hedging activity associated with them. **Dealers hedging these options are forced to unwind their hedge by the end of the trading session because the options expire the same day.** In other words, by construction, **the delta accumulated during the day is a round trip that must be closed before the market closes.**

If dealers are net short gamma around the expiring strikes, they hedge pro-cyclically, selling into declines and buying into rallies. This naturally reinforces intraday trends.

By late afternoon, after a strong trend day, dealers may hold a hedge whose direction is opposite to the day's move. **Unwinding that hedge before settlement can push prices against the prevailing trend. This mechanism could therefore explain both features shown in Figure 2: the same hedging activity that supports intraday trend continuation may also create selling or buying pressure near the close as positions are unwound.**

Of course, this is a simplified explanation. The strongest assumption is that dealers are net short gamma. If they are instead net long gamma during a trend day, the same hedging mechanics could reinforce the trend into the close rather than reverse it. Price data alone cannot answer this question. Testing the hypothesis properly would require strike-level open interest and trade-side classification to estimate dealer gamma throughout the day. Unfortunately, we do not currently have access to these expensive datasets.

Another possible explanation, **more closely related to the general weakening of intraday trends in recent years, is the way information now spreads through the market.** Over the last couple of years, we

have repeatedly seen **markets react aggressively to headlines**, only for those **headlines to be revised, denied, or reinterpreted a few hours later. This is not an ideal environment for intraday trend following.** When information is noisy, incomplete, or sometimes **conveniently timed**, price moves tend to fade more quickly.

Both explanations can be tested. A natural next step would be to separate long and short trend trades and then study the late-day reversal effect conditional on option-market variables, ODTE activity, realized volatility, macroeconomic announcement days, and headline-driven sessions.

Conclusion & Key Takeaways

Over the last few years, we have documented extensively the existence of intraday trends in equity markets. This observation was in clear contrast with the common belief that the lower the time frame, the higher the probability that markets exhibit mean-reverting behavior; the data simply pointed in the opposite direction, at least for the S&P 500.

What we hadn't studied enough was the intraday seasonality of those trends.

In this study, **we found that over the last 20 years, most of the intraday trend PnL was generated during the last hour of the trading session.**

Nevertheless, the evidence presented in this article suggests that **intraday trend seasonality has weakened since 2023.**

In fact, in the recent sample, the late-day contribution has become less reliable, with the final 15 minutes showing a clear tendency for trends to reverse. The ensemble still makes money during the afternoon, but the

final minutes often behave more like a reversal window than a continuation window.

This does not mean that intraday trend following is broken. It means that the **time-of-day structure of the edge may have changed**.

So how should a systematic trader incorporate these findings?

In our opinion, whenever we deal with convex-payoff strategies (and intraday trend following is definitely one of them), adding regime filters that completely switch off trend positions is always risky. **The risk of missing a sudden, unexpected end-of-day rally is not compensated by the benefit of avoiding some reversals.** As a consequence, **we tend to prefer a more conservative approach: reducing part of the trend exposure 15 minutes before the close while keeping some exposure alive to participate in the outsized moves that, sooner or later, will materialize.**

The natural follow-up to this article is to study whether the recent late-day reversal is stronger after positive intraday trends or after negative ones, or whether dealers' positioning could influence the nature of trends during the last part of the session.

If the answer points toward a structural link with ODTE hedging, then late-day exit rules may need to become more adaptive.

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